

Assignment 10: How much for that car?

Zeynep Cakez

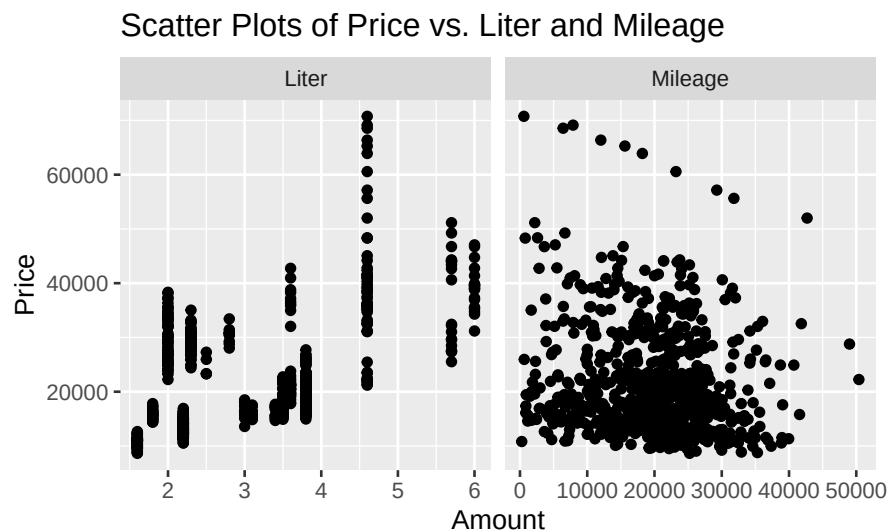
2024-11-03

Exercise 1

- i. There are 2 continuous variables in this dataset. Price is one of them. What is the other?

Mileage

```
car_prices %>%  
  pivot_longer(cols = c(Liter, Mileage), names_to = "PM", values_to = "Amount") %>%  
  ggplot() +  
  geom_point(aes(x = Amount, y = Price)) +  
  facet_wrap(~PM, scales = "free_x") +  
  labs(title="Scatter Plots of Price vs. Liter and Mileage")
```



Exercise 2

```
continuous_model <- lm(Price ~ Liter + Mileage, data = car_prices)  
continuous_model %>%  
  glance() %>%  
  select(r.squared)
```

r.squared
0.3291279

The R^2 of 0.329, Liter and Mileage explain about 32.9% of the variance in car prices, meaning this model has a moderate fit.

Exercise 3

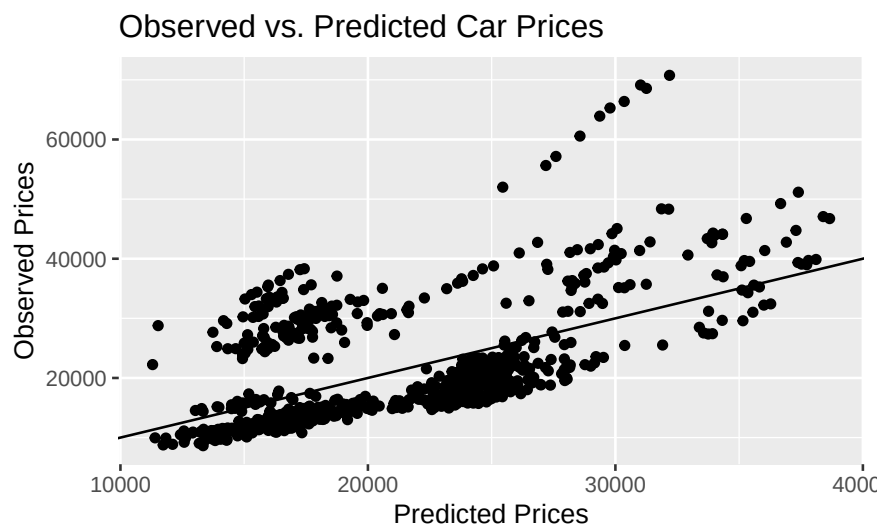
The dots are far from the yellow surface, so that while there is some linearity, but the model doesn't perfectly capture the relationship between price, mileage, and liters. Each liter level shows a line of dots, so the price changes might be linear with those ranges. It's probably easier to see relationships in a 2D model since it simplifies the view, because the 3D model gives more information so it can be harder to interpret.

Exercise 4

```
continuous_df <- car_prices %>%  
  add_predictions(continuous_model, var = "pred") %>%  
  add_residuals(continuous_model, var = "resid")
```

Exercise 5

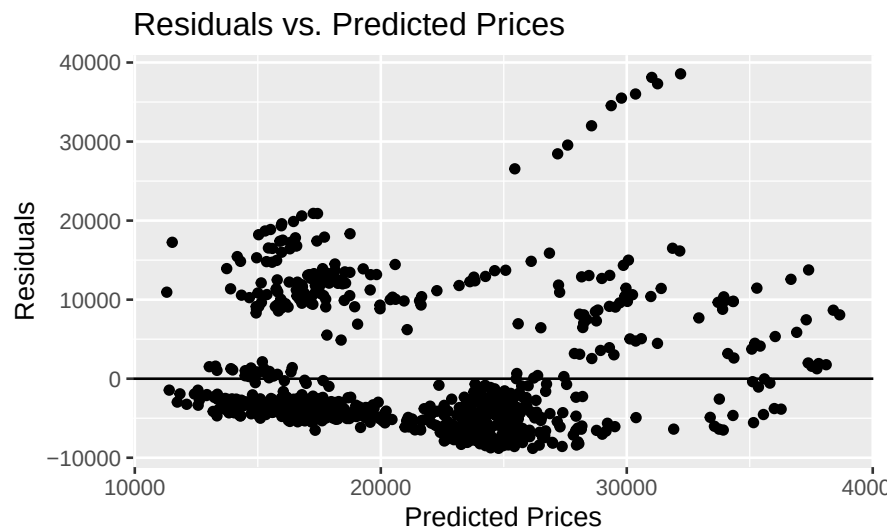
```
continuous_df %>%  
  ggplot() +  
  geom_point(aes(x = pred, y = Price))+  
  geom_abline(slope = 1, intercept = 0)+  
  labs(  
    title = "Observed vs. Predicted Car Prices",  
    x = "Predicted Prices",  
    y = "Observed Prices")
```



The points mostly align with the line, this shows that the model is capturing the overall relationship well, supporting the assumption of linearity. However, their distance from the line indicates that the model may have prediction errors, this mean it could be improved to better fit the actual data.

Exercise 6

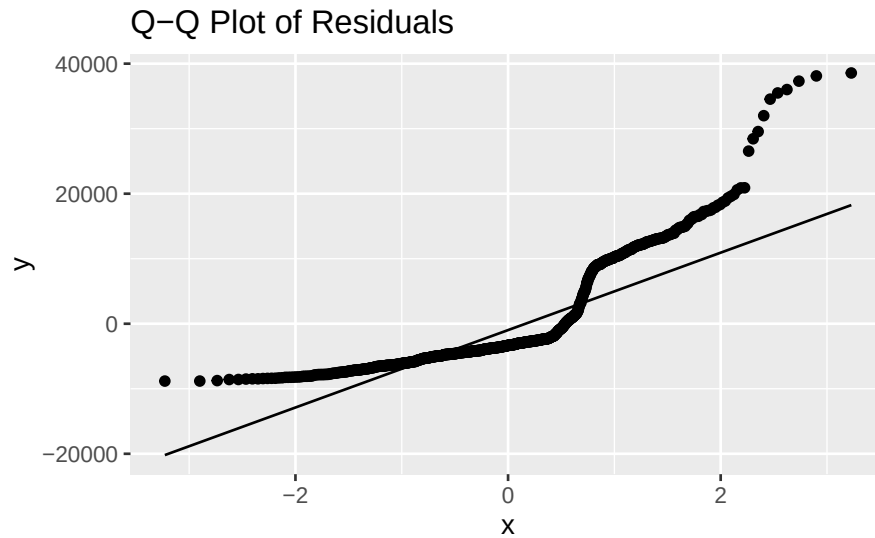
```
continuous_df %>%  
  ggplot() +  
  geom_point(aes(x = pred, y = resid)) +  
  geom_hline(yintercept = 0) +  
  labs(  
    title = "Residuals vs. Predicted Prices",  
    x = "Predicted Prices",  
    y = "Residuals")
```



The residuals are randomly scattered around the horizontal line at $y = 0$, this shows that the variability of the residuals is constant across the range of predicted values.

Exercise 7

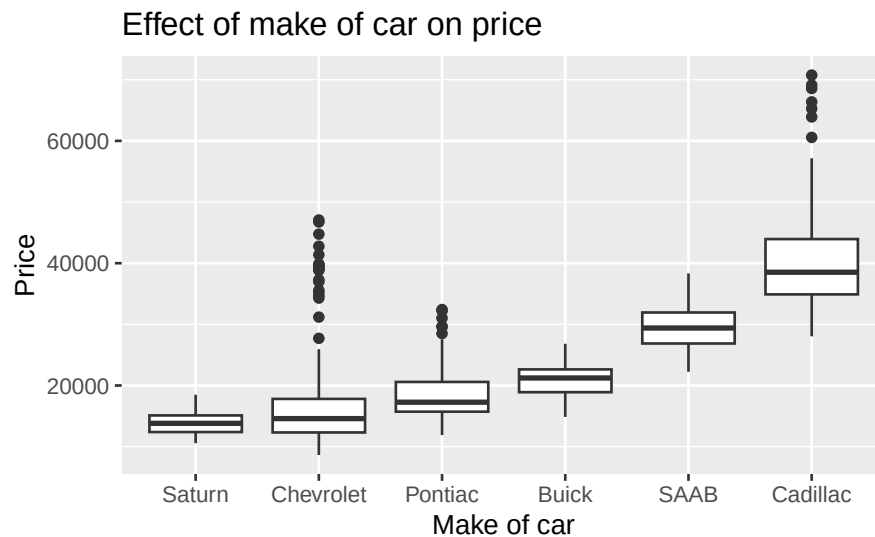
```
continuous_df %>%  
  ggplot() +  
  geom_qq(aes(sample = resid)) +  
  geom_qq_line(aes(sample = resid)) +  
  labs(title = "Q-Q Plot of Residuals")
```



The Q-Q plot shows that the residuals aren't fully normal. They mostly follow a line in the middle but curve away at the ends, which suggests we have some extreme values that wouldn't be there if the data was perfectly normal.

Exercise 8

```
car_prices %>%
  ggplot() +
  geom_boxplot(aes(x = reorder(Make, Price, FUN=median), y = Price)) +
  labs(x = "Make of car", title = "Effect of make of car on price")
```



i. Which make of car has the lowest median price?

Saturn.

ii. Which make of car has the greatest interquartile range of prices?

Cadillac.

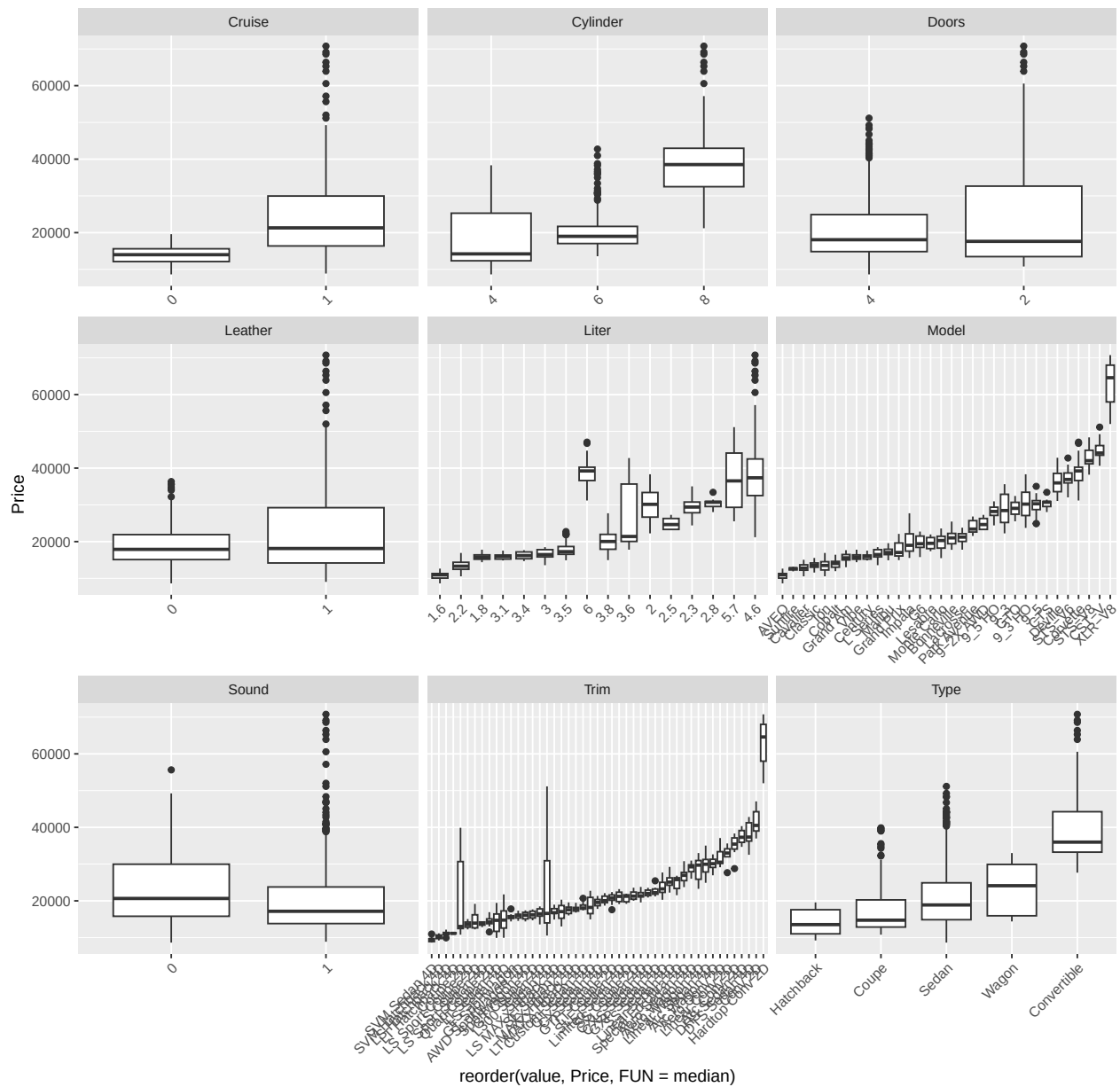
iii. Which makes of cars have outliers?

Chevrolet, Pontiac, and Cadillac.

Exercise 9

```
car_prices %>%
  pivot_longer(
    cols = Model : Leather,
    names_to="name",
    values_to="value",
    values_transform = list(value = 'factor'))%>%
  ggplot() +
  geom_boxplot(aes(x = reorder(value, Price, FUN=median), y = Price))+
  facet_wrap(~name, scales = "free_x")+
  theme(axis.text.x = element_text(angle = 45, hjust = 1))+
  labs(title = "Categorical Variables vs. Price")
```

Categorical Variables vs. Price



Exercise 10

```
cars_factor_df <- car_prices %>%
  mutate(Cylinder = as.factor(Cylinder))

mixed_model <- lm(Price ~ Mileage + Liter + Cylinder + Make + Type, data = cars_factor_df)

mixed_model %>%
  tidy()
```

term	estimate	std.error	statistic	p.value
(Intercept)	1.885018e+04	892.4119413	21.122738	0.0000000
Mileage	-1.861764e-01	0.0106433	-17.492387	0.0000000
Liter	5.697442e+03	342.7322419	16.623596	0.0000000
Cylinder6	-3.312544e+03	619.9683651	-5.343086	0.0000001
Cylinder8	-3.672597e+03	1246.2162662	-2.946998	0.0033032
MakeCadillac	1.450444e+04	517.9855224	28.001635	0.0000000
MakeChevrolet	-2.270807e+03	355.9736337	-6.379145	0.0000000
MakePontiac	-2.355468e+03	363.9063301	-6.472731	0.0000000
MakeSAAB	9.905074e+03	450.2011112	22.001443	0.0000000
MakeSaturn	-2.090266e+03	470.8305609	-4.439529	0.0000103
TypeCoupe	-1.163869e+04	464.7055454	-25.045297	0.0000000
TypeHatchback	-1.172638e+04	545.3936364	-21.500769	0.0000000
TypeSedan	-1.178618e+04	411.1021489	-28.669707	0.0000000
TypeWagon	-8.156551e+03	500.6379995	-16.292312	0.0000000

```
mixed_model %>%
  glance() %>%
  select(r.squared)
```

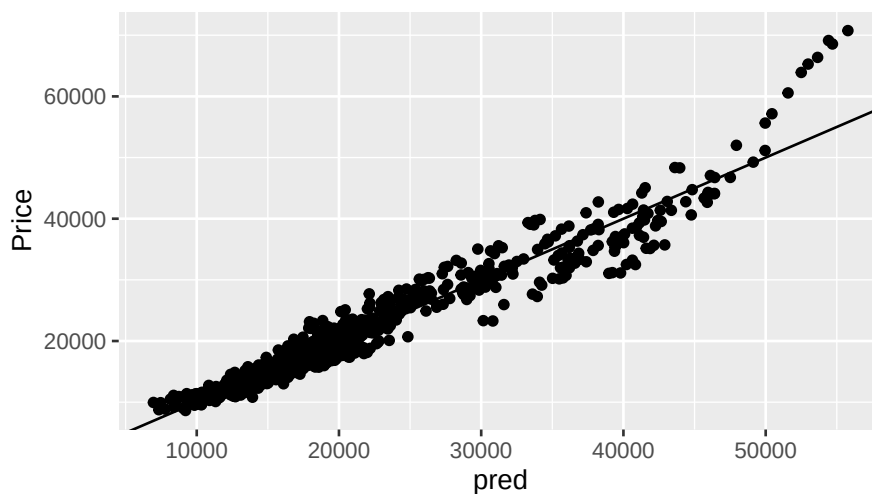
<u>r.squared</u>
0.9389165

Exercise 11

```
mixed_df <- cars_factor_df %>%
  add_predictions(mixed_model) %>%
  add_residuals(mixed_model)

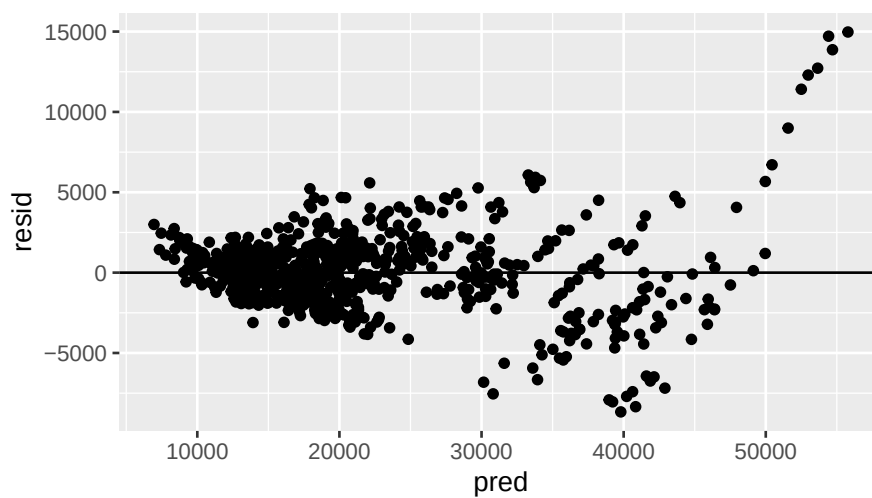
mixed_df %>%
  ggplot() +
  geom_point(aes(x = pred, y = Price)) +
  geom_abline(slope = 1, intercept = 0) +
  labs(title = "Observed vs. Predicted Prices")
```

Observed vs. Predicted Prices

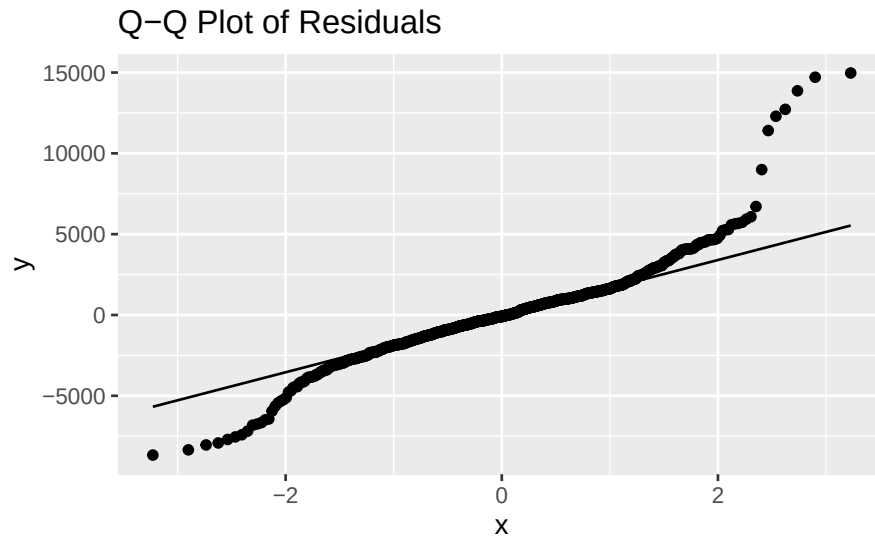


```
mixed_df %>%
  ggplot() +
  geom_point(aes(x = pred, y = resid)) +
  geom_hline(yintercept = 0) +
  labs(title = "Residuals vs. Predicted Prices")
```

Residuals vs. Predicted Prices



```
mixed_df %>%
  ggplot() +
  geom_qq(aes(sample = resid)) +
  geom_qq_line(aes(sample = resid)) +
  labs(title = "Q-Q Plot of Residuals")
```

Exercise 12

- i. How does this mixed model compare to the simpler 2 variable model that we created earlier in the assignment? Consider both how well each model explains variation in Price (R^2) and whether either model violates any of the linear model's three assumptions. Refer back to your graphs and calculations as necessary.

The mixed model is way better at predicting car prices than the simpler model. With an R -squared of 0.94, it explains around 94% of the variation in price, compared to only 33% for the simpler model. Both models meet the key assumptions, but the mixed model is a closer fit, so the points are less scattered in the residual plots.

- ii. Which model would you prefer to use if you were picking a car? Justify your conclusion using your answers to part (i) of this exercise.

If I were picking a car, I would use the mixed model. It gives more accurate predictions on price, it also includes more relevant details about the car.

Academic Integrity statement

No tools other than the teachers guidance from Slack was used to complete this assignment.